

Sadeep Rathnayaka

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About Me

An enthusiastic Electronics and Telecommunication Engineering undergraduate with a passion for software development, robotics and machine learning. I hope to apply my technical knowledge and problem-solving skills to real-world challenges and contribute to the development of innovative solutions.

Education

University of Moratuwa

June 2021 – Present

B.Sc. (Hons) Electronic and Telecommunication Engineering (Undergraduate)

- **GPA: 3.83/4.0** (up to 7th semester)
- **Dean's List: Semeseter 1,4,6**

Nalanda College, Colombo

Jan. 2017 - Dec. 2019

G.C.E Advanced Level Examination 2019 (Physical Science Stream)

- **Z-Score 2.4172 (3As** - Combined Mathematics, Physics and Chemistry)
- Ranked **246th** in the country , **80th** in Colombo District

Work Experience

Industrial Training ([verify](#) [🔗](#))

Dec. 2023 - May.2024

*Robotics Research Internship at **Singapore University of Technology and Design**, supervised by Prof. Tan U-Xuan*

- Developed a robotic system for hospital cleaning using an industrial manipulator integrated with the **Robot Operating System (ROS2)**
- Integrate the manipulator with **Moveit** framework for controlling and trajectory planning
- Designed and implemented a complete simulation environment using **Gazebo** and **RViz** to validate and test the trajectory planning algorithms
- Configured and tested the **Universal Robot UR10e** industrial manipulator using ROS2 to ensure compatibility with the simulation

Projects

POS System (Microservices Architecture)

[🔗](#) [🔗](#) March 2025

(Java | Spring Boot | Spring Cloud | MySQL | WebClient)

- Developed a scalable Spring Boot backend for a Point of Sales(POS) system using microservices.
- Implemented **Role-Based Access Control (RBAC)** and **JWT authentication** at the **API Gateway** to secure service endpoints.
- Integrated **Eureka Server** for dynamic service discovery and load balancing.
- Implemented **synchronous inter-service communication** using **Spring WebClient** for interactions between services.

Inventory Manager Back End (Microservices Architecture)

[🔗](#) [🔗](#) Feb. 2025

(Java | Spring Boot | Spring Cloud |MySQL)

- Developed a microservices-based Spring Boot backend for an inventory system, integrating Spring Cloud for **service discovery** and **API Gateway**.
- Implemented **JWT-based authentication** with a security filter chain for secure access across services.

- Integrated **Eureka Server** for handling multiple micro-service instances and load balancing.

Inventory Manager Back End (Monolithic Architecture)

  Jan. 2025

(*Java* | *Spring Boot* | *MySQL*)

- Developed a scalable Spring Boot backend for an inventory system with CRUD operations.
- Designed and implemented the database schema with **multiple interconnected tables**, ensuring proper relationships, data integrity, and normalization.
- Optimized **custom query methods** for inventory search with pagination.

EcoEx Mini Game Back End (Monolithic Architecture)

  Nov. 2023 – May 2024

(*Java* | *Spring Boot* | *MySQL*)

- Implemented **JWT authentication** for secure player access and stateless sessions.
- Designed MySQL database schema with relational tables for multiple game entities.
- Optimized inventory and player data handling with CRUD operations and pagination.

Flower Exchange Project by London Stock Exchange Group

  Sep. 2023

(*C++* | *OOP*)

- Designed a program that can be used as a flower trading platform using OOP concepts

Mobile Robot Receptionist: A Semi Humanoid Robot

  Aug. 2024 – Present

(*ROS2* | *Moveit* | *C++* | *Python* | *Solidworks*)

- Implemented the 3D model for 4 degrees-of-freedom robotic arm
- Controlled a robotic arm system using **ROS2** and **Moveit** frameworks
- Implemented ROS2 nodes using C++ and Python to control the robotic arm

Parasitic Egg Detection and Classification

  Oct. 2023

(*Image Processing* | *Machine Learning* | *Python*)

- The RTM-Det-tiny model was used for object detection, optimizing accuracy by slicing larger images into patches, performing inference on them, and then recombining them for final results

Achievements

RoboRoaz Competition 2024 - Singapore ([verify](#))

- **2nd Runners Up** - Reconfigurable Robot Navigation and Algorithm building in a simulated environment

Robotic Design Competition - Semester 3, 2023 (Virtual)

- **Champions** - Simulations in Webots

SparkLink 1.0 - 2023 ([verify](#))

- **Champions** - Hardware Description Language competition organized by University of Ruhuna

Skills

Programming Languages: Java, C++, Python

Web Development: HTML, CSS

Frameworks : Spring Boot, ROS2, Moveit, Nav2

Databases: MySQL

Tools: Git, Postman, Swagger, Solidworks, Altium

Online Courses

Udemy : Robotics and ROS 2 - Learn by Doing! Manipulators ([verify](#) )

Cousera : Supervised Machine Learning: Regression & Classification ([verify](#) )

Cousera : Advanced Learning Algorithms ([verify](#) )

Cousera : Unsupervised Learning, Recommenders, Reinforcement Learning ([verify](#) )

Cousera : Introduction to Computer Vision and Image Processing ([verify](#) )

References

Dr. Peshala Jayasekara
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Commendation from Dr. Oh Xueyan 